ENVS 193DS - Spring Final

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Github Repo

My repo here

Set Up

Load packages

```
# load in packages
library (tidyverse)
library (here)
library (janitor)
library (rstatix)
library (ggeffects)
library (scales)
library (MuMIn)
library (DHARMa)
```

Load data

```
# create object for nest data
nests <- read.csv (here("data", "nest_data_final.csv"))
# create object for bike commute data
bike_commute <- read.csv(here("data", "ENVS_193DS_Personal_Data.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In **part 1**, my coworker used a linear model (linear regression) to perform their analysis.The response variable is binary and reflects the probability that the American Avocet (*Recurvirostra americana*) uses the mirco habit (1) or doesn't (0). The predictor is a continuous variable describing the distance to water edge (m).

In **part 2**, they used a chi-squared test for analysis to determine whether there is a relationship between two categorical variables. There categorical variables are bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat type (managed pond, non-tidal marsh, salt pond, tidal marsh, tidal mudflat).

b. Suggestions for rewriting

Part 1: The data suggests that American Avocets are more likely to use micro habitats that are closer to the water's edge. We can reject our null hypothesis because the data shows a strong relationship that is unlikely due to random chance.

Part 2: The three shorebird species used the five habitat types in unequal proportions. The species showed a clear pattern of habitat preference which are unlikely to appear solely from random chance.

c. Further analysis

Another variable that could influence the probability of American Avocet micro habitat use, is Salinity (ppt) which is a continuous variable that can be measured with a salinity meter. It might have an impact on micro-habitat use because salinity influences the composition and abundance of aquatic invertebrates which serve as prey for American Avocets and therefore could predict Avocet distribution.

Problem 2. Data analysis

a. Clean and wrangle

```r
nests_clean <- nests |>
  # select columns of interest
  select (height_cm, case_control, ant_species) |>
  # filter to only include ant species C. mimosae, C. sjostedti,
  # and C. nigriceps
  filter (ant_species %in% c ("AB", "RRB", "BBR")) |>
  # relevel ant species to be in the following order: 1) C. sjostedti,
  # 2) C. mimosae, 3) C. nigriceps
  mutate (ant_species = fct_relevel (ant_species, "AB", "RRB", "BBR")) |>
  # create a new column to store the full scientific name for each ant species
  mutate (scientific_name = case_when (
      ant_species == "AB"  ~ "Crematogaster sjostedti",
      ant_species == "RRB" ~ "Crematogaster mimosae",
      ant_species == "BBR" ~ "Crematogaster nigriceps"))

# structuring clean data
str (nests_clean)
```

```
'data.frame':   270 obs. of  4 variables:
 $ height_cm      : num  392 310 513 154 324 ...
 $ case_control   : int  1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ scientific_name: chr  "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mimos
```

```r
# display 10 observations from nest_clean
slice_sample(nests_clean, n = 10)
```

```
   height_cm case_control ant_species         scientific_name
1      103.0            0         RRB   Crematogaster mimosae
2      174.5            1         RRB   Crematogaster mimosae
3      132.0            0         RRB   Crematogaster mimosae
4      341.0            0         RRB   Crematogaster mimosae
5      390.0            1         BBR Crematogaster nigriceps
6      324.5            0          AB Crematogaster sjostedti
7       74.0            0         RRB   Crematogaster mimosae
8      129.5            0          AB Crematogaster sjostedti
9       99.0            0         RRB   Crematogaster mimosae
10     121.5            0         RRB   Crematogaster mimosae
```

b. Response variable

The response variable case_control is binomial and looks at whether a tree contained a nest or if it was was an "available" tree. If the tree contained a nest, it recieved a "1", and if the tree did not contain a nest, it recieved a "0".

c. Purpose of study

- Tree height is a continuous predictor variable that is measured in cm. As tree height increases, the probability of nest occurrence most likely increases because the birds will be able to nest farther away from potential predators on the ground (Results, paragraph 2).
- Acacia ant species is a categorical predictor variable with 3 levels: *Crematogaster sjostedti* (AB), *Crematogaster mimosae* (RRB), and *Crematogaster nigriceps* (BBR). It could influence the probabilty of birds nesting in the tree because some ant species are symbiont ants tolerating nesting birds as well as other predators, and other ant species can be hostile to "browsing animals" and predators (introduction, paragraph 2-3).

d. Table of models

Model number	Tree Height (cm)	Ant Species	Model description
1	X	X	All predictors (No interaction)
2	X	X	All predictors (Interaction term)

e. Run the models

```
# model 1: all predictors, no interaction
model1 <- glm (
  # formula: case_control as a function of tree height + ant species
  case_control ~ height_cm + ant_species,
  data = nests_clean, # data = nests_clean
  family = binomial # response variable is binomial
)

# model 2: all predictors, interaction term
model2 <- glm (
  # formula: case_control as a function of tree height * ant species
  case_control ~ height_cm * ant_species,
```

```
  data = nests_clean, # data = nests_clean
  family = binomial # response variable is binomial
)
```

f. Select the best model

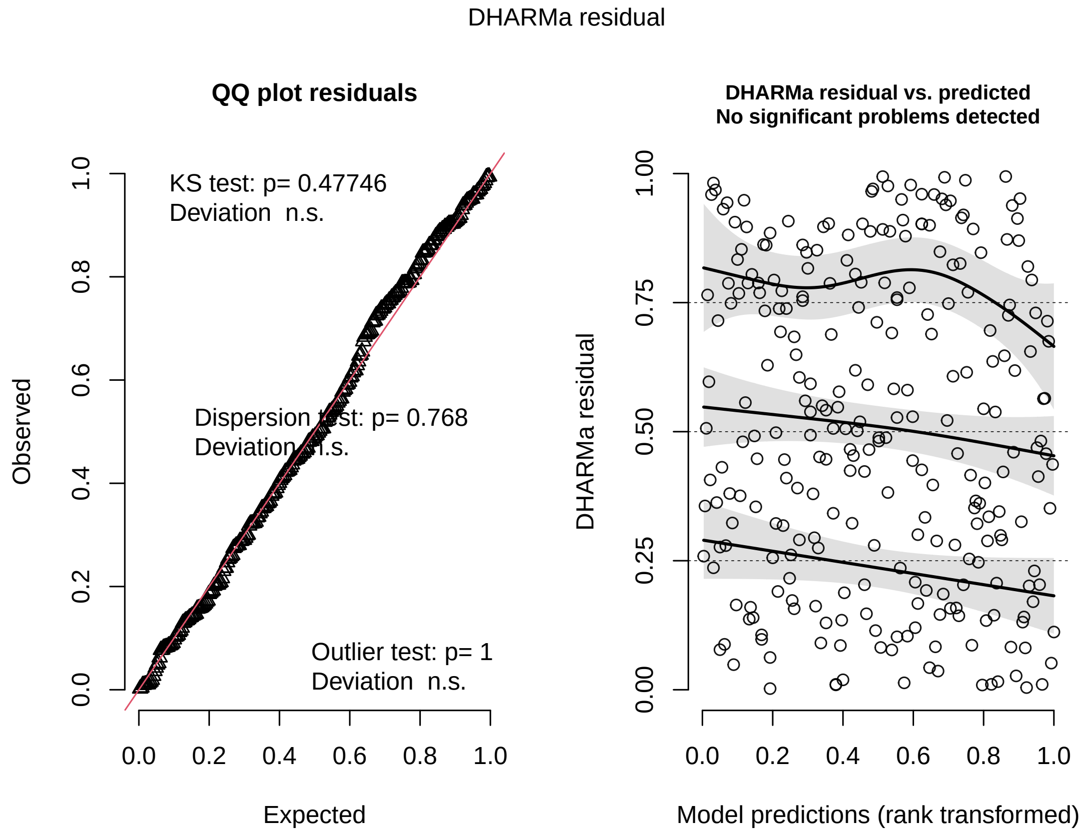
```
# calculating AIC
AICc (
  model1,
  model2
) |>
  # arranging output in order of AIC
  arrange (AICc)
```

```
       df     AICc
model2  6 238.1236
model1  4 258.8940
```

The best model as determined by Akaike's Information Criterion (AIC) is model 2. This model includes "case_control" as a function of all predictors (tree height and ant species) and uses an interaction term. This model explains the most variation in the simplest way.

g. Check the diagnostics

```
# simulated residuals from GLM for diagnostic plots
plot(simulateResiduals(model2))
```



Information from left plot (QQ plot residuals):

- KS test: Residuals follow a uniform distribution.
- Dispersion test: Residuals are not more dispersed than expected from a uniform distribution.
- Outlier test: There are not outliers in the data set that could be influencing model coefficients or predictions.

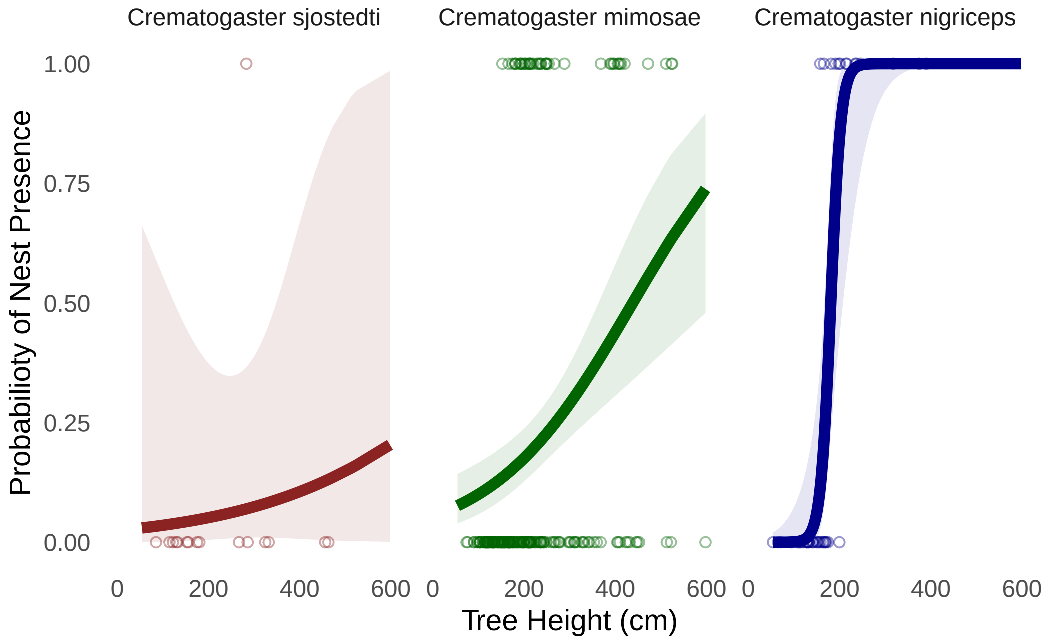
Information from right plot (residuals vs predicted values): There are randomly arranged residuals across the range of predicted values. There are no significant problems detected.

h. Visualize the model predictions

```r
# new object called nests preds
nests_preds <- ggpredict (model2, # model2 object
                          # prediction variables
                          terms = c("height_cm [all]", "ant_species")) |>
  rename(height_cm = x,
         ant_species = group)
```

```r
# baselayer: ggplot
# data: nests clean
ggplot( data = nests_clean,
        mapping = aes (x = height_cm, # x-axis = height
                       y = case_control)) + # y-axis = case_control
  # first layer: geom_point
  # Create point for individual binary observations
  geom_point (aes (color = ant_species), # color by ant_species
              shape = 21, # open circles
              alpha = 0.4) + # making circles transparent
  # second layer: geom_line
  # create line for predicted probability of nest occurence
  geom_line (data = nests_preds, # data used is nests predictions
             # line x = height_cm, y = predicted value
             mapping = aes (x = height_cm,
                  y = predicted,
                  color = ant_species), # colored by ant species
             # made line wider
             linewidth = 2) +
  # third layer: geom_ribbon
  # create 95% CI interval
  geom_ribbon (data = nests_preds, # data used is nests predictions
              mapping = aes (
                    y = predicted,
                    ymin = conf.low, # bottom of CI interval
                    ymax = conf.high, # top of CI interval
                    fill = ant_species), # fill with ant_species data
               alpha = 0.1) + # make ribbon transparent
  # created panels for each ant species
  facet_wrap (~ ant_species,
              # labelling each panel by species name
              labeller = labeller (ant_species = c(
               "AB" =  "Crematogaster sjostedti",
               "RRB" = "Crematogaster mimosae",
               "BBR" = "Crematogaster nigriceps")
```

```
            )) +
  # changed panels to custom colors for lines and points
  scale_color_manual (values = c(
    "AB" = "brown4",
    "RRB" = "darkgreen",
    "BBR" = "darkblue"
  )) +
  # changed to custom colors for ribbons
  scale_fill_manual (values = c (
    "AB" = "brown4",
    "RRB" = "darkgreen",
    "BBR" = "darkblue"
  )) +
  # set x axis range and breaks
  scale_x_continuous (limits = c (0, 600),
                      breaks = c (0, 200, 400, 600)) +
  # relable x and y axis
  labs (x = "Tree Height (cm)",
        y = "Probabilioty of Nest Presence") +
  # changed theme
  theme_minimal() +
  # remove grid lines
  theme (panel.grid = element_blank ()) +
  # removed legend
  theme (legend.position = "none")
```



i. Write a caption for your figure.

Figure 1: The probability of a bird nest occurring in a tree increases as tree height increases across all 3 ant species presence. The figure visualizes how the probability of nest presence (y-axis) is predicted by tree height in "cm" (x-axis) interacting with ant species (seperate panels). Open cirlces represent individual observation of binary presence or absence of tree nests on trees of a certain height, seperated by ant species. Line in each panel describes model prediction of nest occurrence probability as tree height changes and ribbon describes the 95% confidence interval of this prediction. Colors correspond to ant species, Red panel: "*Crematogaster sjostedti*", green panel: "*Crematogaster mimosae*", blue panel: "*Crematogaster nigriceps*". Data sourced from: Cre Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: *Symbiotic acacia ants drive nesting behavior by birds in an African savanna* [Data set]. Zenodo. https://doi.org/10.5281/zenodo.8373322

j. Calculate model predictions

```
# what is the probability of nest occurrence at a value of 600 cm for each
# ant_species
ggpredict(model2,
terms = c("height_cm [600]", "ant_species"))
```

```
# Predicted probabilities of case_control

ant_species: AB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

ant_species: RRB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species: BBR

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your results

The predicted probability of nest site occupancy at 600 cm (the tallest tree) is less for *Crematogaster sjostedti* (0.20 (95% CI [0.00, 0.99])) and *Crematogaster mimosae* (0.74 (95% CI [0.48, 0.90])) than for *Crematogaster nigriceps* (1.00 (95% CI [1.00, 1.00])). There is a 100% probability that nests will occur in trees at height 600 cm with the ant species *Crematogaster nigriceps* present. Additionally, the probability of a bird nest occurring in a tree increases as tree height increases across all 3 ant species presence. Birds had a higher probability to nest in trees inhabited by aggressive defenders of host trees (*C.nigriceps* and *C.mimosae*), but rarely nested in trees that are inhabited by less aggressive defenders (*C.sjostedti*) (Figure 1). Nesting sites that are inhabited by more aggressive ant species can reduce the risk of nest predation (Lujan et al., 2023, Young et al., 1990).

Problem 3. Affective and exploratory visualizations

a. Comparing visualizations

- My affective visualization does not include each individual observation as a data point like my exploratory data visualization does from homework 2. Instead, my affective visualization uses a bike path visual that represent the trend of the data and includes bikers overlaid to represent clusters of data points along the line.

- Both visualizations exhibit a clear positive linear shape that can be used to infer the relationship I discovered between bike commute duration and commute urgency.
- Between both visualizations, there is a clear relationship between bike commute duration and commute urgency. The less time I have before class when I leave home, the quicker I bike from home to campus.
- One piece of helpful feedback I received in workshop was that the way I represented outliers with the same object as my trendline was distracting and confusing, and I used this suggestion to create a more cohesive visualization by changing this representation. I also was recommended to better differentiate the bikers on the bottom left of my trendline to better represent that they are moving faster than other "points". I utilized this suggestion by adding dust clouds behind the "faster" bikers.

b. Designing an analysis

One statistical analysis that I could use to answer my research question is a linear model (linear regression). Because my central question of my data collection is; "How does the amount of time I leave my house before class effect the duration of my bike commute?", I can use a linear model to display this predicative relationship. My predictor variable is how much time I have from when I leave home until class start (minutes), and my response variable is the duration of bike commute (minutes). Both of these variables are continuous. I would expect these two variable to be connected and observe that my urgency to get to class predicts how quickly I ride my bike.

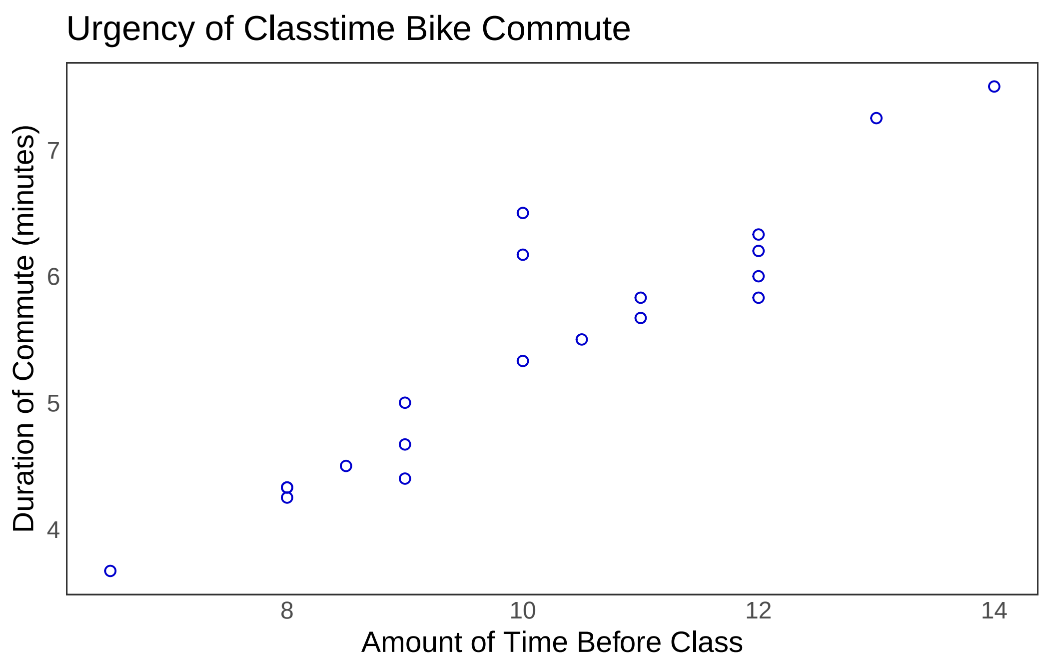
c. Check any assumptions and run your analysis

Clean data

```r
# create new object for clean data
bike_commute_clean <- bike_commute |>
  # clean column names
  clean_names() |>
  # select columns of interest
  select(duration_m,
         time_before_class_min)
```

Visualize data

```
# base layer: ggplot
# data frame: bike_commute_clean
ggplot (data = bike_commute_clean,
      mapping = aes(
      # x-axis: amount of time before class
      # y-axis: duration of commute
      x = time_before_class_min,
      y = duration_m)) +
  # first layer: point = scatterplot
  # make data points colored, change shape
  geom_point(color = "blue3",
             shape = 21) +
  # Changed title, x and y-axis
  labs (title = "Urgency of Classtime Bike Commute",
        x = "Amount of Time Before Class",
        y = "Duration of Commute (minutes)") +
  # applied custom plot theme
  theme_bw() +
  # removing the panel background
  theme(panel.background = element_blank()) +
  # removing the panel grid
  theme(panel.grid = element_blank()) +
  # removing axis ticks
  theme(axis.ticks = element_blank())
```



Independent observations

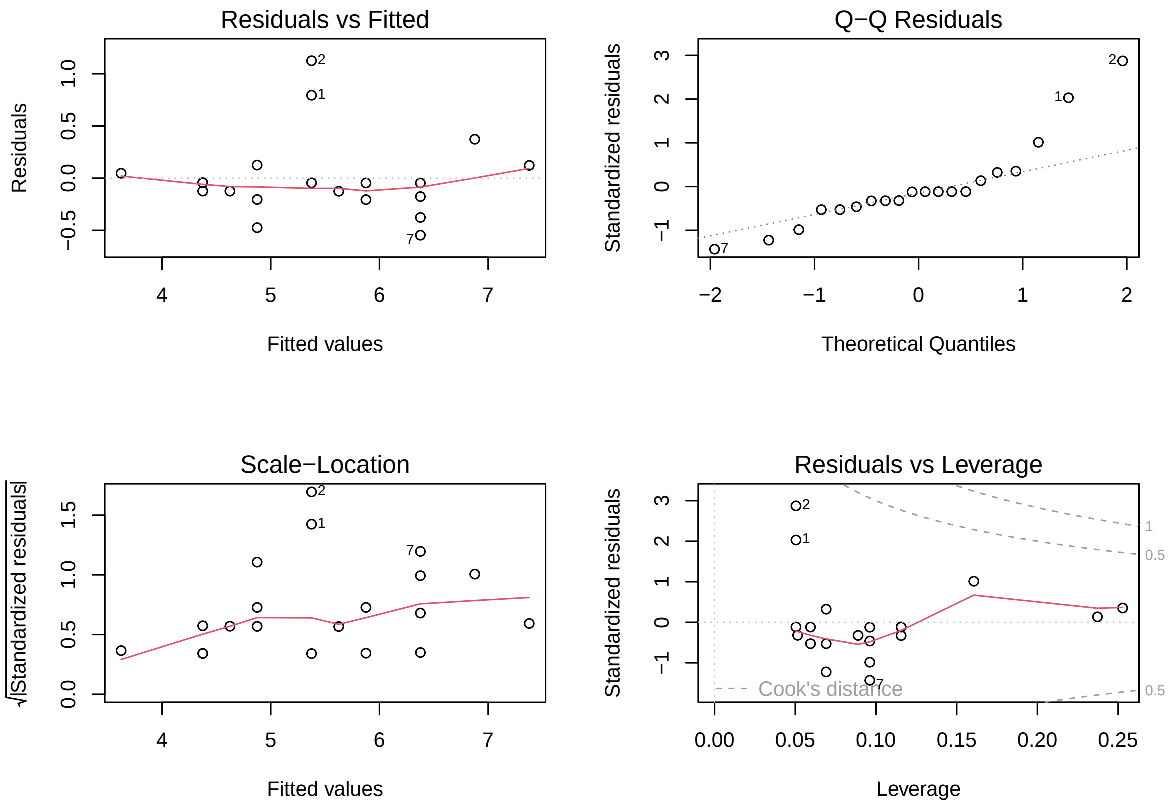
My data has independent observations. Each individual observation is independent of the observations before that. My bike commute duration to class one day, does not effect my next commute duration.

Write out model

```
# create new object for bike commute
# duration as a function of time before class
bike_commute_model <- lm (duration_m ~ time_before_class_min,
                          data = bike_commute_clean) # data: bike_commute_clean
```

Check assumptions with diagnostic plots

```
# display diagnostic plots in 2 by 2 grid
par (mfrow = c(2, 2))
# plot bike commute model
plot (bike_commute_model)
```



Note: Using the "Residuals vs Fitted" and "Scale-Location" plots, we can determine that the residuals are evenly distributed and have constant variance (homoscedasticity). Using the "QQ Residuals" plot we can determine that the residuals are normally distributed. Finally, using the "Residual vs Leverage" plot, we can determine that there are no outliers that are influencing our model.

All assumptions are met

Model summary

```
#bike commute model summary
summary (bike_commute_model)
```

```
Call:
lm(formula = duration_m ~ time_before_class_min, data = bike_commute_clean)
```

```
Residuals:
     Min       1Q   Median       3Q      Max 
-0.54666 -0.18368 -0.04634  0.06564  1.12461 

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)    
(Intercept)            0.36905    0.48971   0.754    0.461    
time_before_class_min  0.50063    0.04731  10.582 3.72e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4018 on 18 degrees of freedom
Multiple R-squared:  0.8615,	Adjusted R-squared:  0.8538 
F-statistic:   112 on 1 and 18 DF,  p-value: 3.716e-09
```

Note:

Intercept: 0.36905 ± 0.4897

Slope: 0.50063 ± 0.0473

Adjusted R-squared: 0.8538

P-value: $p < 0.0001$

d. Create a visualization

Model predictions

```r
# new object call bike_commute_preds
bike_commute_preds <- ggpredict (bike_commute_model, # model object
                                 terms = "time_before_class_min") # predictor

# display the predictions
bike_commute_preds
```

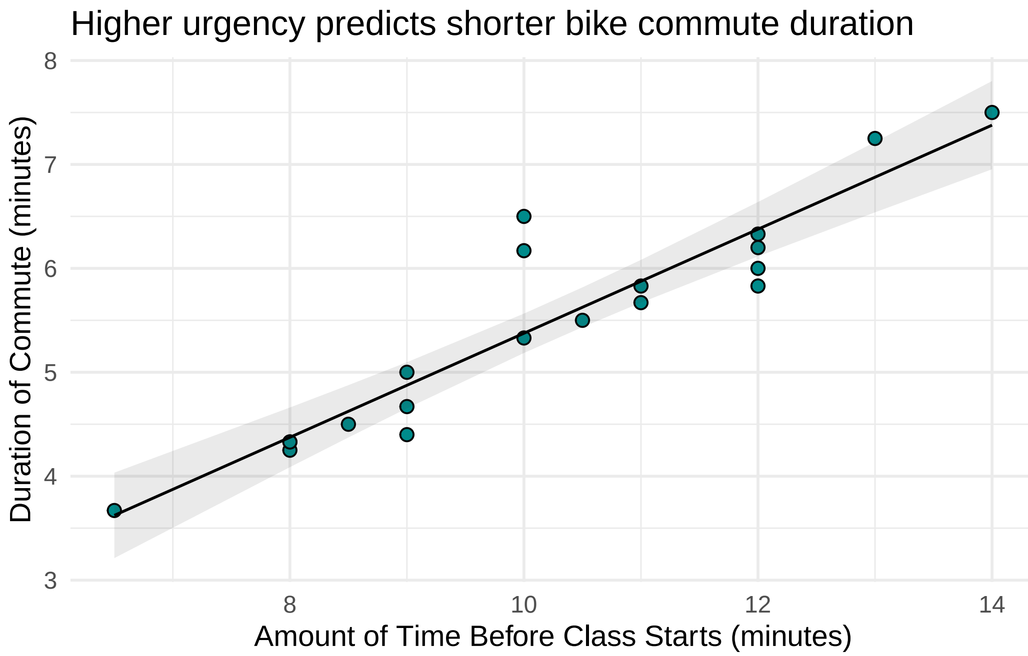
```
# Predicted values of duration_m

time_before_class_min | Predicted |     95% CI
-----------------------------------------------
                 6.50 |      3.62 | 3.21, 4.03
```

```
   8.00 |       4.37 | 4.09, 4.66
   8.50 |       4.62 | 4.37, 4.88
   9.00 |       4.87 | 4.65, 5.10
  10.50 |       5.63 | 5.43, 5.82
  11.00 |       5.88 | 5.67, 6.08
  12.00 |       6.38 | 6.11, 6.64
  14.00 |       7.38 | 6.95, 7.80
```

Visualization

```
# base layer: ggplot
# using clean data frame
ggplot(data = bike_commute_clean,
       aes(x = time_before_class_min,
           y = duration_m)) +
  # first layer: points representing individual observations
  geom_point(size = 2,
             stroke = 0.5,
             fill = "cyan4",
             shape = 21) +
  # second layer: ribbon representing confidence interval
  # using predictions data frame
  geom_ribbon(data = bike_commute_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              alpha = 0.1) +
  # third layer: line representing model predictions
  # using predictions data frame
  geom_line(data = bike_commute_preds,
            aes(x = x,
                y = predicted)) +
  # axis labels
  labs( title = "Higher urgency predicts shorter bike commute duration",
       x = "Amount of Time Before Class Starts (minutes)",
       y = "Duration of Commute (minutes)") +
  # theme
  theme_minimal()
```



e. Write a caption

Figure 2: Leaving home with less time before class predicts a shorter bike commute Visualization displays linear relationship between the "Amount of Time Before Class Starts (minutes)" on the x-axis, and the "Duration of Commute (minutes)" on the y-axis. Blue points represent individual bike commute observations. Line represents model predictions, and ribbon displays 95% confidence interval.

f. Write about your results

My biking commute duration increased as the amount of time I had before class increased (linear regression, $F(1, 18) = 112$, $p < 0.0001$, $\alpha = 0.05$). 85% ($R^2 = 0.85$) of the variation in the model can be explained the predictor.

Based on model predictions, for every 1 minute increase in available time before class starts, I expect an increase in commute duration of 0.5 ± 0.047 (SE) minutes.

g. Sharing your affective visualization

I attended week 10 workshop! :)